

ECO 362 Financial Investments

Precept 5: Exam Review & CAPM

October 20 - 27, 2025

Exam Review: General Comments

- We want to give you partial credit:
 - Never leave anything blank.
 - Show us the steps! An incorrect final numerical answer will get 0 points by itself, but may earn partial credit for the derivation.
 - Exception: don't spam formulas in hopes of partial credit
 - If you submit multiple answers for the same question, your score will be the one earned by the worse answer.
- The structure of your answer matters:
 - 1) Establish what data you have, and what you want to find.
 - 2) Find a formula for what you want to find in terms of what you have. Use as many intermediate steps as you need. E.g. ZCB prices in this class
 - 3) Show how you insert the data into the formula.
 - 4) Calculate a numerical answer.
- Too many careless mistakes! Make sure you correctly:
 - Convert between percentages and numerical values, i.e. $0.1\% = 0.001$.
 - enter formulas into your calculator.

See Solutions.pdf

- Capital asset pricing model (CAPM): asset pricing in general equilibrium
 - Supply, demand, and market clearing
- Key assumptions:
 - Investors have identical beliefs about distribution of returns
 - Investors are all mean-variance expected utility optimizers

- What is a market portfolio?
 - $\mathbf{w}_m$: market weights on risky assets
 - The i -th element: asset i 's market equity as a share of total market equity
 - $1 - \mathbf{w}_m' \mathbf{1}$: market weight on riskless asset
- The return on market portfolio:

$$r_m = r_b + \mathbf{w}_m'(\mathbf{r} - r_b \mathbf{1})$$

- As $\mathbb{E}[r_m] = r_b + \mathbf{w}_m'(\mathbb{E}[\mathbf{r}] - r_b \mathbf{1})$, we know:

$$r_m - \mathbb{E}[r_m] = \mathbf{w}_m'(\mathbf{r} - \mathbb{E}[\mathbf{r}]) = \mathbf{w}_m'(\mathbf{r} - \boldsymbol{\mu})$$

- Supply of risky assets in dollars: $A\mathbf{w}_m$, with A being total wealth in the economy

- Recall the optimal portfolio choice of investor n is:

$$\mathbf{w}_n = \frac{1}{\gamma_n} \Sigma^{-1} (\boldsymbol{\mu} - r_b \mathbf{1}) = \frac{1}{\gamma_n} \Sigma^{-1} \mathbb{E}[\mathbf{r} - r_b \mathbf{1}]$$

- Demand of investor n for risky assets in dollars: $A_n \mathbf{w}_n$
- Aggregate demand of all investors for risky assets in dollars: $\sum_{n=1}^N A_n \mathbf{w}_n$, where A_n is the wealth of investor n .

- Supply = Demand

$$A\mathbf{w}_m = \sum_{n=1}^N A_n \mathbf{w}_n = \sum_{n=1}^N \frac{A_n}{\gamma_n} \Sigma^{-1} \mathbb{E}[\mathbf{r} - r_b \mathbf{1}]$$

$$\mathbf{w}_m = \sum_{n=1}^N \frac{A_n}{A\gamma_n} \Sigma^{-1} \mathbb{E}[\mathbf{r} - r_b \mathbf{1}]$$

- Denote $\frac{1}{\bar{\gamma}} = \sum_{n=1}^N \frac{A_n}{A\gamma_n}$, then

$$\mathbf{w}_m = \frac{1}{\bar{\gamma}} \Sigma^{-1} \mathbb{E}[\mathbf{r} - r_b \mathbf{1}]$$

- Rewrite as: (Both sides $I \times 1$ vectors)

$$\mathbb{E}[\mathbf{r} - r_b \mathbf{1}] = \bar{\gamma} \Sigma \mathbf{w}_m = \bar{\gamma} \mathbb{E}[(\mathbf{r} - \boldsymbol{\mu})(\mathbf{r} - \boldsymbol{\mu})' \mathbf{w}_m]$$

- Recall $r_m - \mathbb{E}[r_m] = (\mathbf{r} - \boldsymbol{\mu})' \mathbf{w}_m$, then

$$\mathbb{E}[\mathbf{r} - r_b \mathbf{1}] = \bar{\gamma} \mathbb{E}[(\mathbf{r} - \boldsymbol{\mu})(r_m - \mathbb{E}[r_m])] \quad (1)$$

- The i -th row of this equation is:

$$\mathbb{E}[r_i - r_b] = \bar{\gamma} \mathbb{E}[(r_i - \mathbb{E}[r_i])(r_m - \mathbb{E}[r_m])] = \bar{\gamma} \text{Cov}(r_i, r_m)$$

$$\mathbb{E}[r_i - r_b] = \bar{\gamma} \text{Cov}(r_i, r_m)$$

- The last step is to get rid of $\bar{\gamma}$
- Clever idea: Multiply both sides of equation (1) by $\mathbf{w}'_m$:

$$\mathbb{E}[\mathbf{w}'_m(\mathbf{r} - r_b \mathbf{1})] = \bar{\gamma} \mathbb{E}[\mathbf{w}'_m(\mathbf{r} - \mu)(r_m - \mathbb{E}[r_m])]$$

- Recall that : $r_m - r_b = \mathbf{w}'_m(\mathbf{r} - r_b \mathbf{1})$ and $r_m - \mathbb{E}[r_m] = \mathbf{w}'_m(\mathbf{r} - \mu)$

$$\mathbb{E}[r_m - r_b] = \bar{\gamma} \mathbb{E}[(r_m - \mathbb{E}[r_m])^2] = \text{Var}(r_m)$$

- This gives a formula for $\bar{\gamma}$

$$\bar{\gamma} = \frac{1}{\text{Var}(r_m)} \mathbb{E}[r_m - r_b]$$

- CAPM:

$$\mathbb{E}[r_i - r_b] = \frac{\text{Cov}(r_i, r_m)}{\text{Var}(r_m)} \mathbb{E}[r_m - r_b] = \beta_i \mathbb{E}[r_m - r_b]$$

- β_i is asset i 's market beta, which is given by

$$\beta_i \equiv \frac{\text{Cov}(r_i, r_m)}{\text{Var}(r_m)}$$

- The market beta for the market portfolio is 1;
- The market beta for the idiosyncratic risk is 0 and $\mathbb{E}[r_i] = r_b$;

- Assets with higher market beta have higher expected return;
- Idiosyncratic risk (uncorrelated with market return) is **not** rewarded by the market

Application of CAPM: Present-value formula

- An asset that costs P today and pays a random cash flow X at a future date (e.g., in 10 years) and has market beta β_i
- Rate of return (over 10 years, not annualized)

$$1 + r_i = \frac{X_i}{P_i}$$

- Apply the CAPM:

$$\mathbb{E} \left[\frac{X_i}{P_i} - 1 - r_b \right] = \beta_i \mathbb{E}[r_m - r_b]$$

$$P_i = \frac{\mathbb{E}[X_i]}{1 + r_b + \beta_i \mathbb{E}[r_m - r_b]}$$

- Data on risk free rates r_{bt} , market returns r_{mt} , and asset i returns r_{it}
- Estimate linear regression

$$r_{it} - r_{bt} = \alpha_i + \beta_i(r_{mt} - r_{bt}) + \epsilon_{it}$$

- Report estimates $\hat{\alpha}_i, \hat{\beta}_i$ and their standard errors.
- Use t-stat or p-value to determine statistical significance.
- CAPM predicts $\alpha_i = 0$. If $\hat{\alpha}_i$ is statistically significantly different from zero, evidence against CAPM.
- Excel Regression via Data Analysis Tab (install ToolPak as an Add-In)

- Sorting stocks by **beta**; low beta stocks have positive alpha.
- Sorting stocks by **residual variance**; stocks with lowest variance have "abnormally" (according to CAPM) high returns.